Advanced Portfolio Management - Key Concepts for Portfolio X-Ray

Book Overview: "Advanced Portfolio Management: A Quant's Guide for Fundamental Investors" by Giuseppe Paleologo teaches fundamental equity investors how to use quantitative risk models and portfolio construction techniques to separate stock-specific alpha from systematic factor returns, manage risk effectively, and maximize risk-adjusted performance through proper position sizing and hedging.

Chapter 3-4: Factor Model Framework

Master Equation (Section 11.1.1)

```
r_t = α + B_t * f_t + ε_t

Where:
- r_t: n-dimensional vector of asset total returns
- α: n-dimensional vector of expected returns (alphas)
- B_t: n×m matrix of factor loadings
- f_t: m-dimensional vector of factor returns
- ε_t: n-dimensional vector of idiosyncratic returns
```

Asset Covariance Matrix

```
Ω_r = B * Ω_f * B' + Ω_ε

Where:
- Ω_r: Total return covariance
- Ω_f: Factor covariance matrix
- Ω_ε: Idiosyncratic covariance (usually diagonal)
```

Factor Return Estimation (Section 11.1.2)

For fundamental models, estimate factor returns via weighted least squares:

```
min (r - B*f)' W (r - B*f)

Solution: f̂ = (B'WB)^-1 * B'W * r
Residuals: ε̂ = (I - B(B'WB)^-1 B'W) * r

Where W is diagonal with W_ii = (market_cap_i)^0.5
```

Chapter 5: Key Factors

Fama-French 3-Factor Model

1. **Market (MKT)**: Broad market return (SPY or S&P 500)
2. **Size (SMB)**: Small Minus Big - small cap premium
3. **Value (HML)**: High Minus Low - value premium (Book-to-Price)

Additional Factor: Momentum (MOM)

- 12-month momentum: returns over past year excluding most recent month
- Formula: MOM = r(t-12, t-1) where t-1 excludes last month

Factor Loadings

- Market beta: Regression coefficient of stock returns on market returns
- Size: log(market cap) - mean(log(market cap)) / std
- Value: Book-to-Price ratio, z-scored
- Momentum: Past 12-month return, z-scored

Chapter 6-7: Risk Decomposition

Portfolio Factor Exposure (Section 11.1.4)

```
b = B' * w

Where:
- w: portfolio weights (n-dimensional)
- b: factor exposures (m-dimensional)
```

Percentage Idiosyncratic Variance (Section 11.1.3)

```
% idio var = (w' * Ω_ε * w) / (b' * Ω_f * b + w' * Ω_ε * w)

This measures purity of the portfolio:
- 100% = pure stock selection, no factor risk
- <75% = significant factor risk present
```

Marginal Contribution to Factor Risk (Section 11.1.5)

```
MCFR_i = [B * Ω_f * b]_i / sqrt(b' * Ω_f * b)

Used to identify which positions contribute most to factor risk
```

Portfolio Beta to Benchmark (Section 11.1.4)

$\beta(w,v) = (w' * \Omega_r * v) / (v' * \Omega_r * v)$

Where v is the benchmark portfolio weights

Chapter 8: Performance Metrics

Sharpe Ratio (Section 6.1)

Sharpe Ratio = $E[r] / \sigma(r)$

Annualized from daily:
$SR_{annual} = \text{sqrt}(252) * SR_{daily}$

Information Ratio

$IR = E[\varepsilon] / \sigma(\varepsilon)$

Where ε is idiosyncratic return
Measures skill independent of factor exposures

Performance Attribution (Section 8.1.1)

Total PnL = Idio PnL + Factor PnL

Factor PnL = Σ ($exposure_i \times factor\_return_i$)
Idio PnL = $w' \times \varepsilon$

Hit Rate and Diversification (Section 11.2)

$IR = (2p - 1) \times \text{sqrt}(N_{eff}) \times \text{sqrt}(2 \times 252 / \pi)$

Where:
- p: hitting probability (% of correct calls)
- N_{eff}: effective number of stocks = $||w||_1^2 / ||w||_2^2$

Chapter 11: Key Formulas Reference

Factor-Mimicking Portfolios (Section 11.1.2)

```
V = (B'WB)^-1 * B'W

Each column v_i is a portfolio that tracks factor i
These are used for hedging
```

Proportional Rule for Sizing (Section 11.4)

```
Given expected returns α, find factor-neutral positions:
1. Regress: α = B*x + η
2. Residuals η are factor-neutral alphas
3. Position sizing: NMV_i = (GMV / Σ|η_i|) × η_i
```

Risk-Based Sizing Options (Section 6.3)

1. **Proportional**: $NMV_i \propto \alpha_i$
2. **Risk Parity**: $NMV_i \propto \alpha_i / \sigma_i$
3. **Mean-Variance**: $NMV_i \propto \alpha_i / \sigma_i^2$

Book recommends Proportional for most cases.

Implementation Notes

Data Requirements

1. Portfolio holdings: Ticker, Shares, BuyDate, (optional) SellDate
2. Historical prices: Daily adjusted close prices
3. Factor returns: Can use Kenneth French data library or construct from market data

Risk Model Construction Steps

1. Load portfolio and fetch price history
2. Calculate stock returns
3. Fetch/construct factor returns (Market, Size, Value, Momentum)
4. Estimate factor loadings (betas) via regression
5. Calculate factor exposures: $b = B'w$
6. Decompose variance into factor and idio components
7. Calculate performance metrics

Fama-French Factor Construction

- **Market**: SPY or S&P 500 returns
- **Size (SMB)**: Small cap index (IWM) **-** Large cap (SPY)
- **Value (HML)**: Value ETF (IWD) **-** Growth ETF (IWG)
- **Momentum**: Can construct from cross-sectional stock returns

Or download directly from Kenneth French's data library:
https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Key Insights from Book

1. **Separate alpha from beta** (Ch 3): Focus on idiosyncratic returns
2. **Target 75%+ idio variance** (Ch 7): Keep factor risk low
3. **Proportional sizing works best** (Ch 6): Simple beats complex
4. **Diversification is key** (Ch 8): N_eff = 50-200 optimal
5. **Hit rate matters more than sizing** (Ch 8): Focus on selection

Output Requirements

1. Factor exposures (dollar and percentage)
2. % Idiosyncratic variance
3. Sharpe Ratio and Information Ratio
4. Performance attribution (factor vs idio)
5. Cumulative returns decomposition chart
6. Factor exposure bar chart
7. Excel export with all metrics